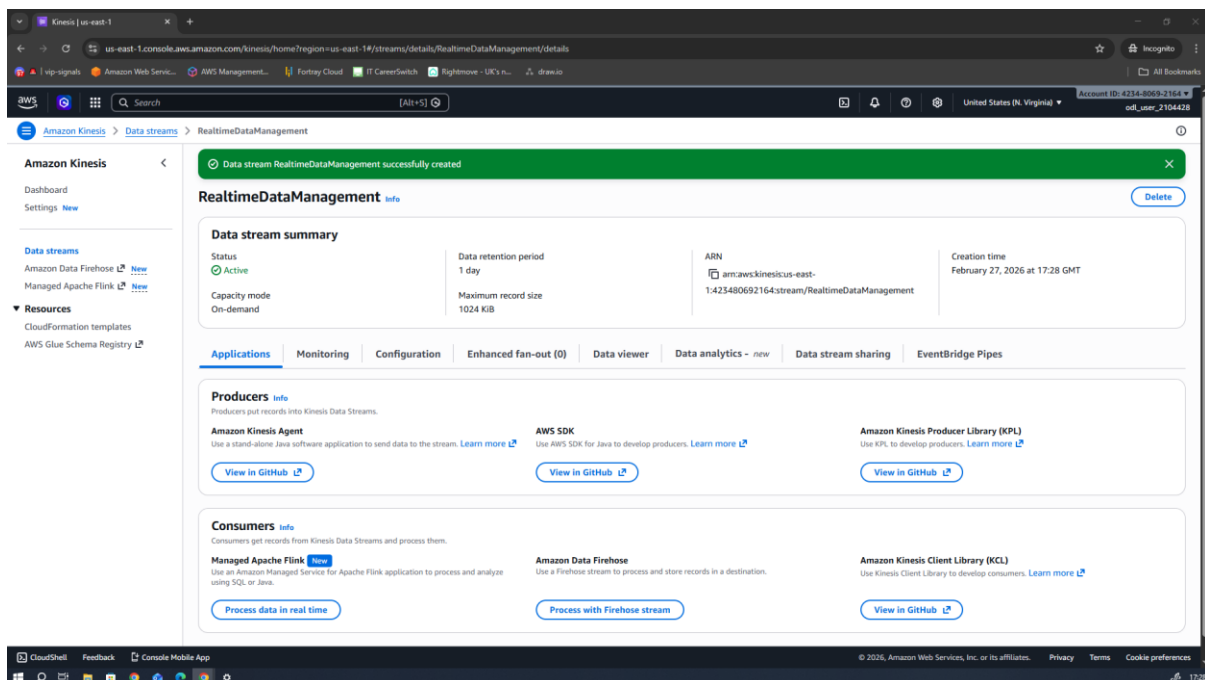
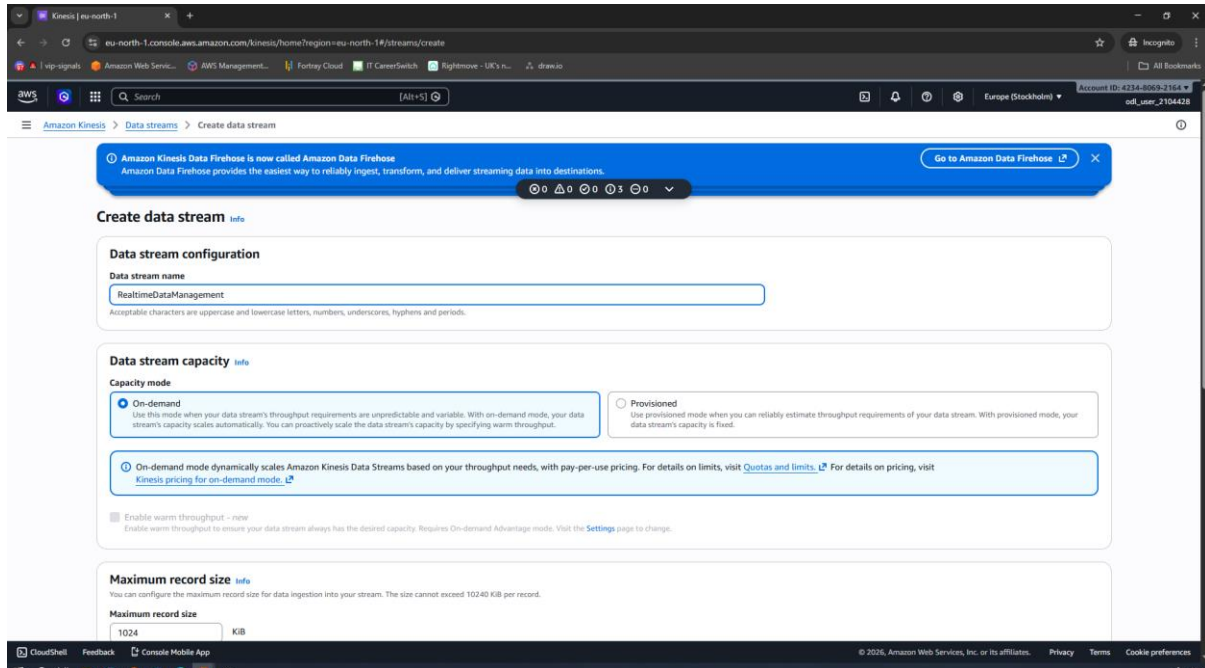
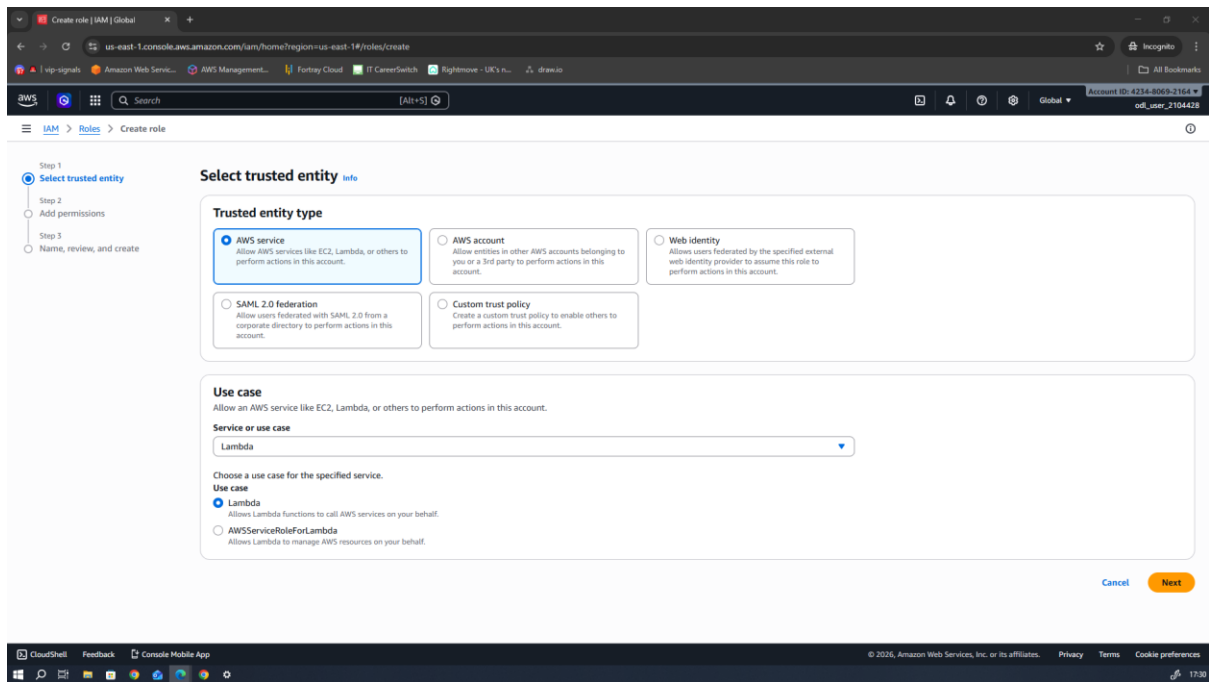


# Project: Real-time Data Management

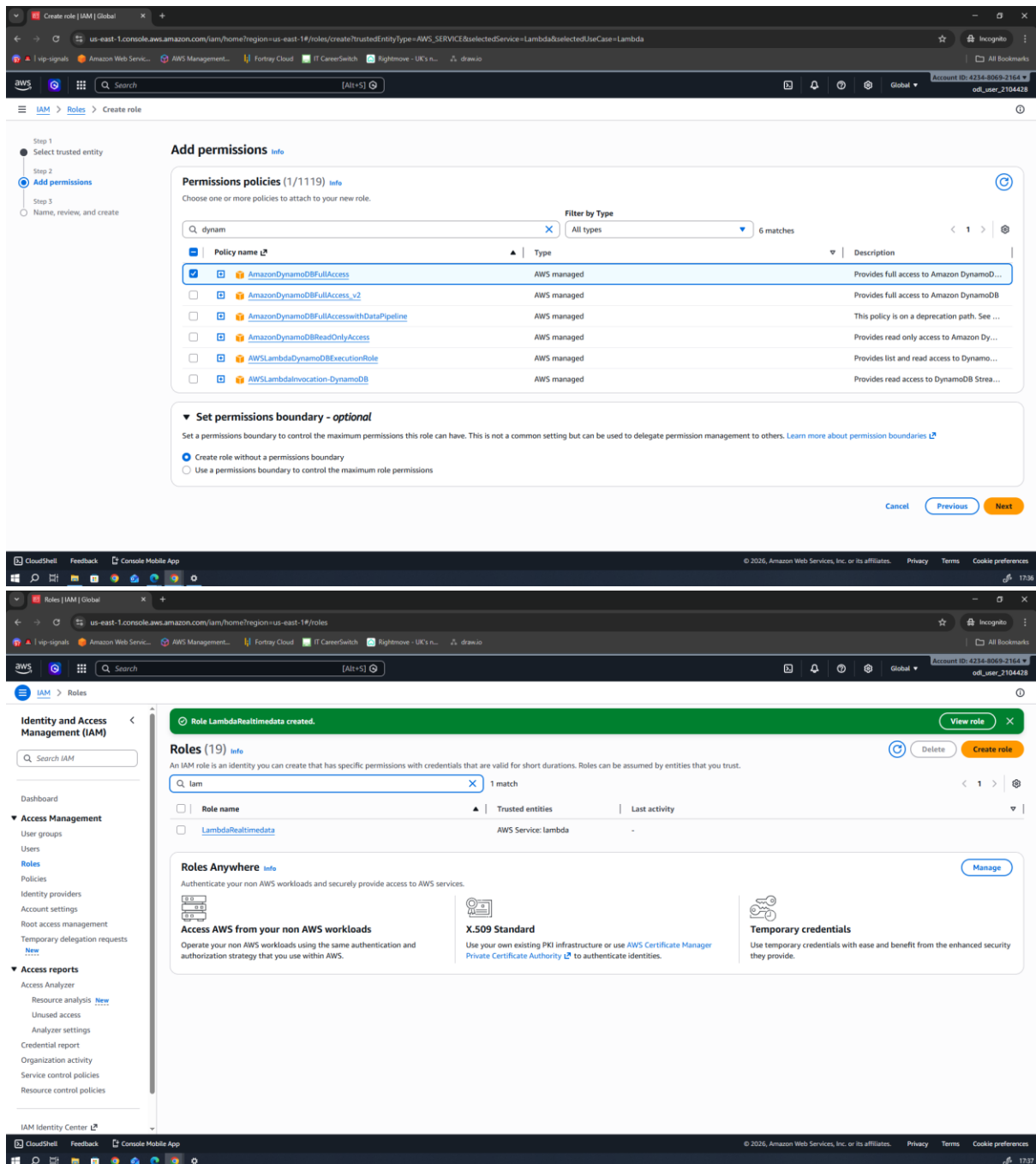
1<sup>st</sup> I created a Kinesis Data stream selected on demand but would choose provisioned if I knew throughput requirements name RealtimeDataManagement



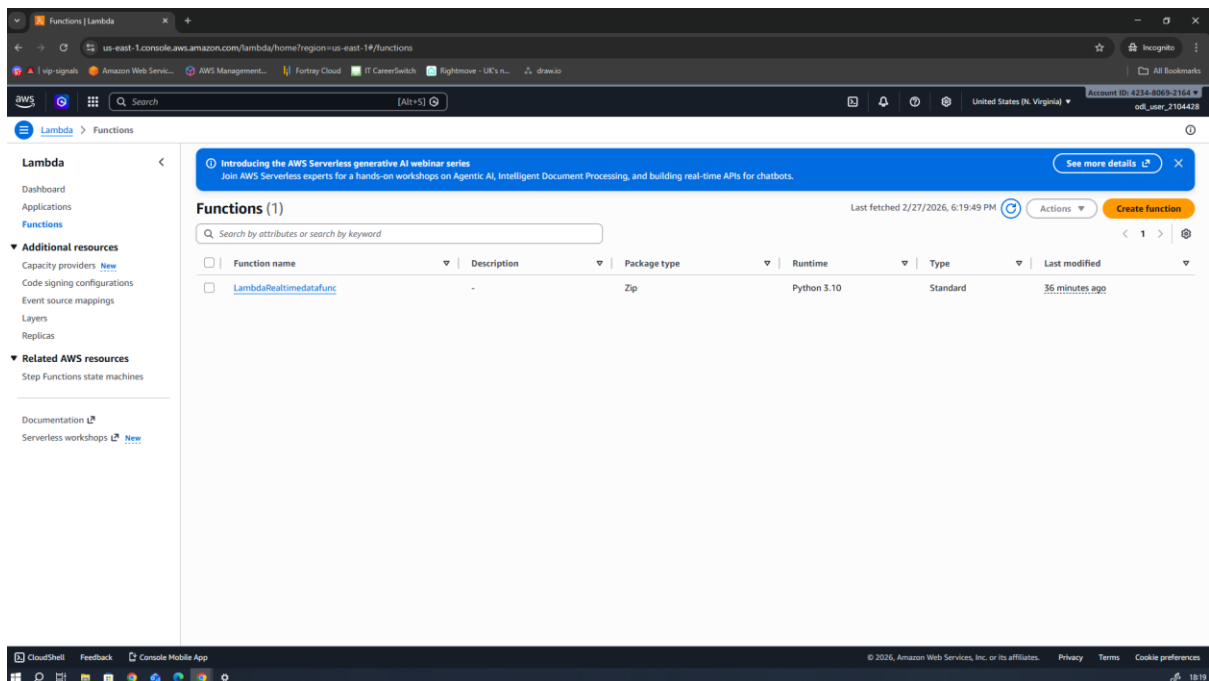
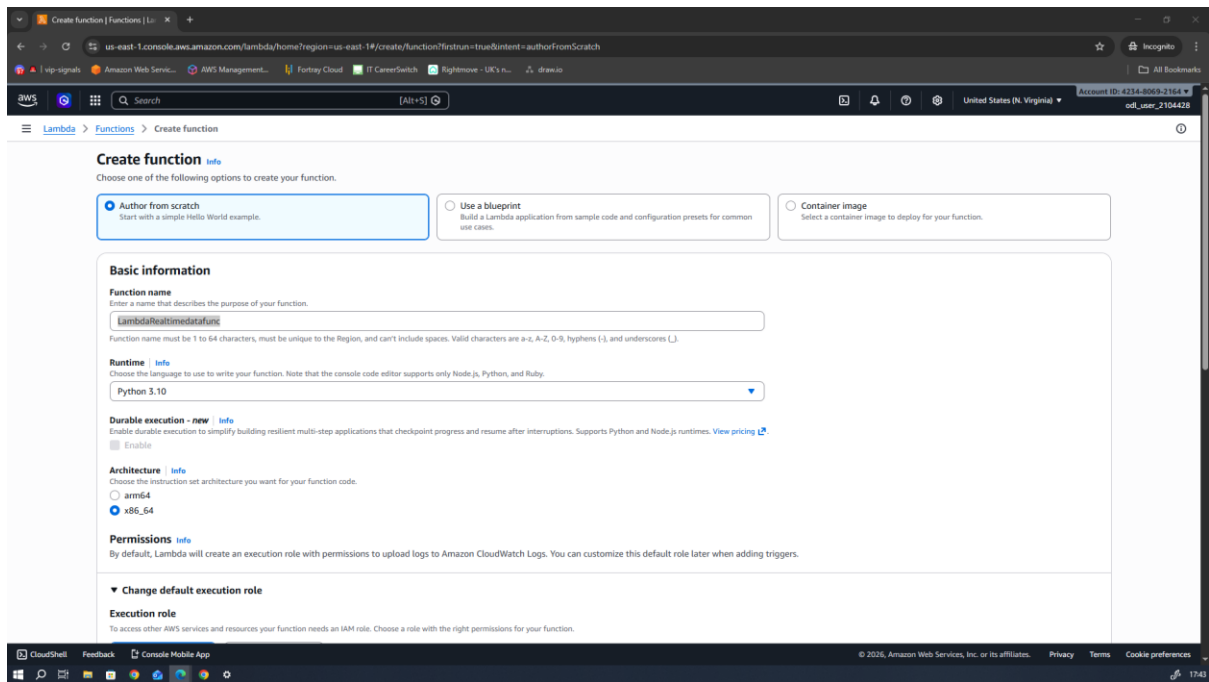
Next I have created a role for Lambda name: LambdaRealtimedata



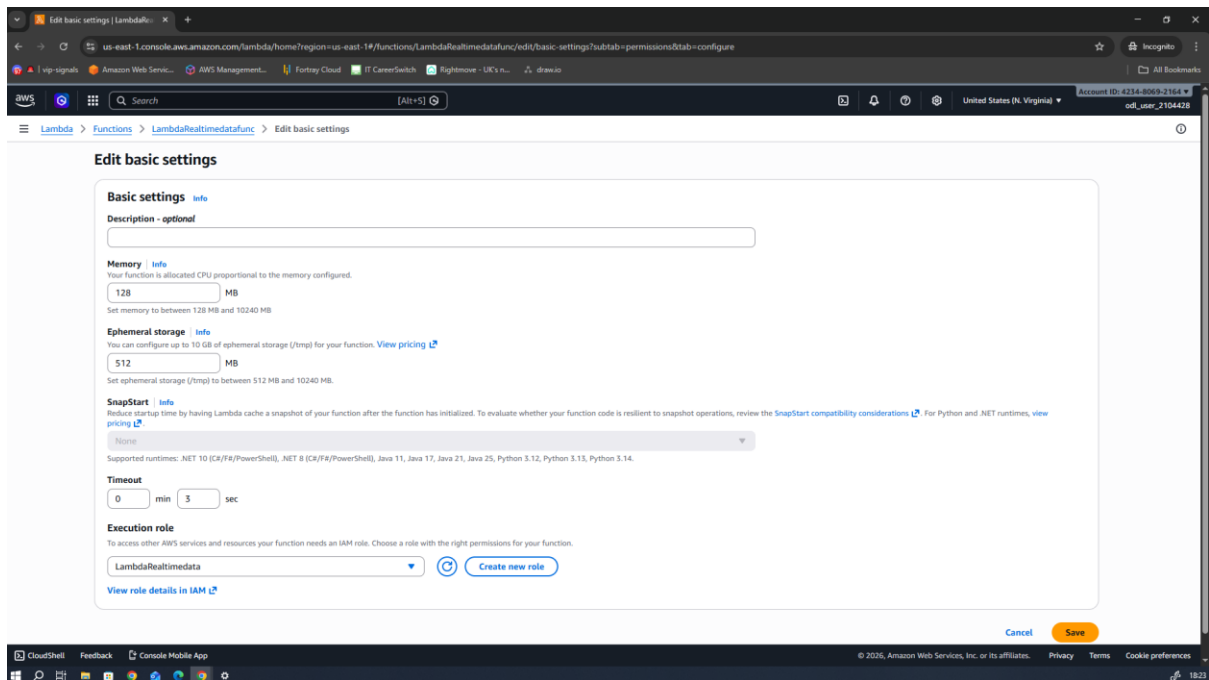
used policy for dynamodb full access



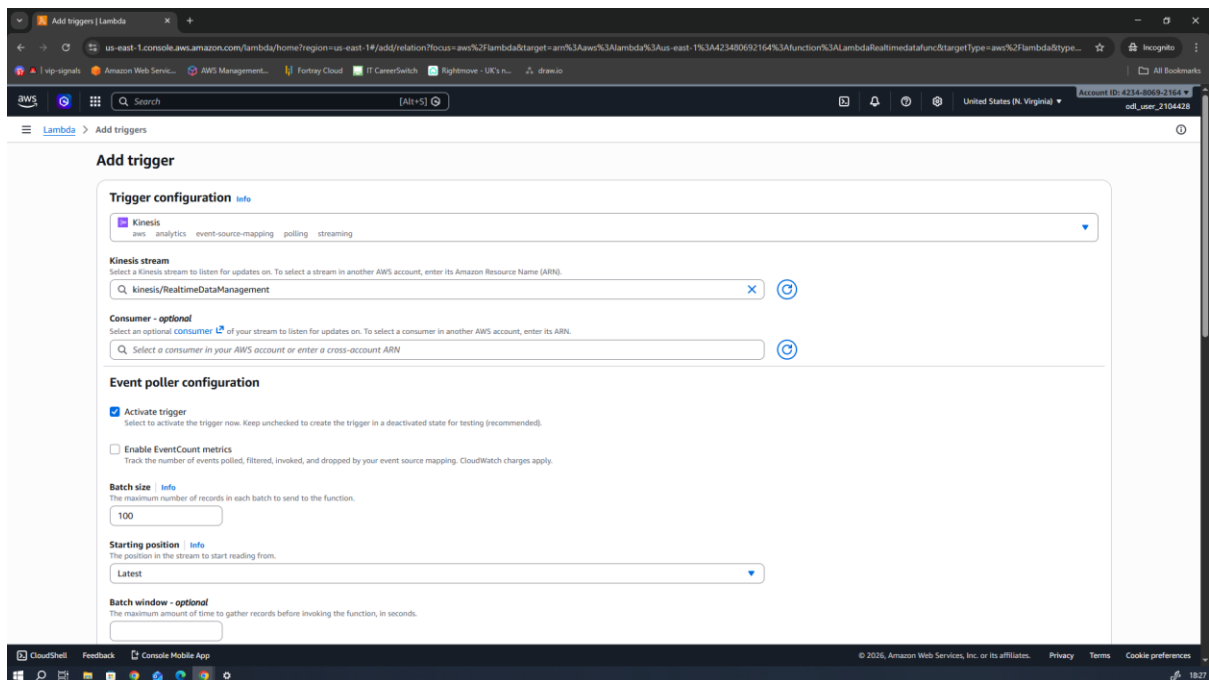
Then I created a Lamda function Name: LambdaRealtimedatafunc



Role LambdaRealtimedata assigned to lambda function



Trigger set to kinesis/RealtimeDataManagement



Created Dynmo DB database named Telemax-DynDB partition key "Telemax-DynDBpk"

Added code below to Lambda function :

```
import json
import base64
import uuid
import boto3
from datetime import datetime

dynamodb = boto3.resource("dynamodb")
table = dynamodb.Table("Telemax-DynDB")

def lambda_handler(event, context):
    results = {
        "records_received": len(event["Records"]),
        "records_written": 0,
        "errors": []
    }

    for record in event["Records"]:
        try:
            # Decode base64 Kinesis payload
            raw_data = base64.b64decode(record["kinesis"]["data"]).decode("utf-8")
            data = json.loads(raw_data)

            # Ensure DynamoDB partition key exists
            if "Telemax-DynDBpk" not in data:
                data["Telemax-DynDBpk"] = str(uuid.uuid4())
```

```
# Optional: add timestamp if useful for your project

if "ingestTimestamp" not in data:

    data["ingestTimestamp"] = datetime.utcnow().isoformat()


# Write to DynamoDB

table.put_item(Item=data)

results["records_written"] += 1

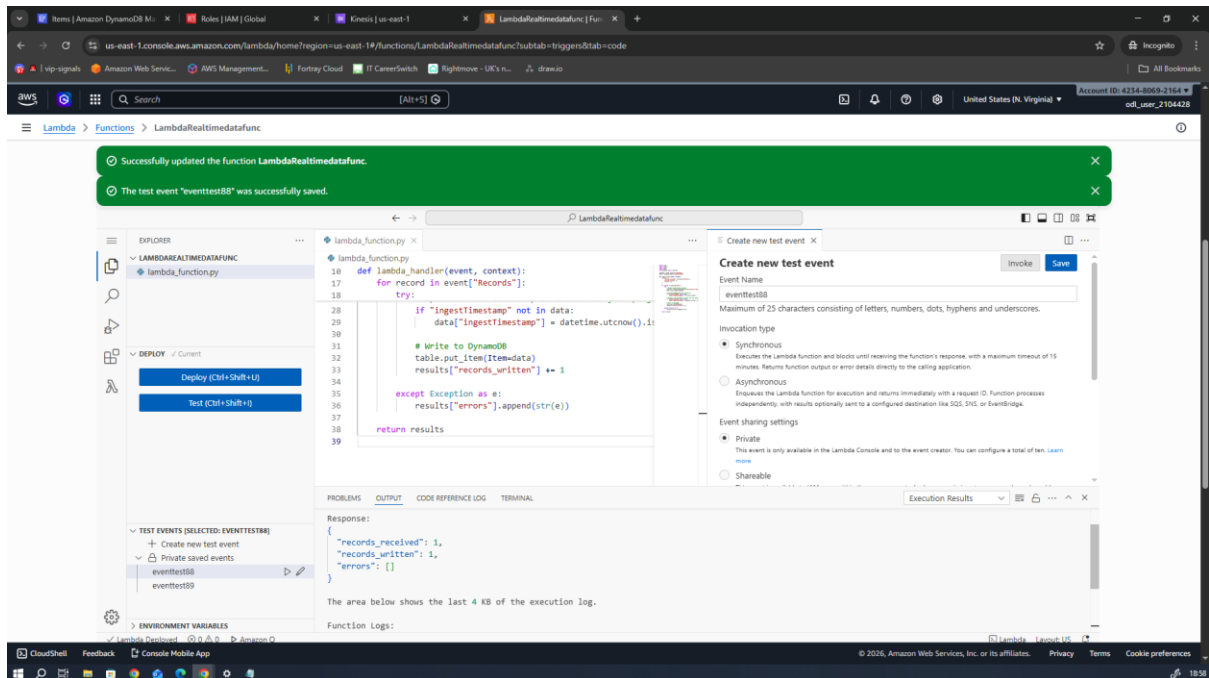

except Exception as e:

    results["errors"].append(str(e))

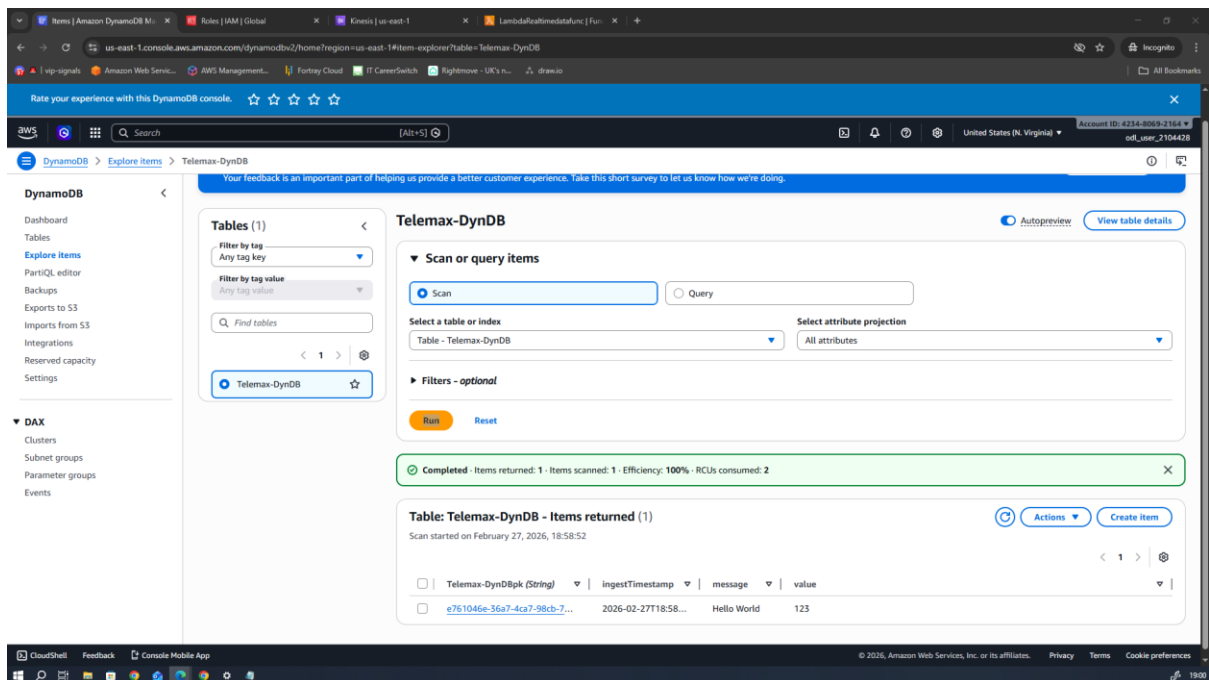

return results
```

This Lambda function takes every record coming from my **Kinesis stream**, decodes it, ensures it has the required **DynamoDB partition key**, adds a timestamp, and then writes it into my **Telemax-DynDB** table.

Event tested by creating new event which looks to have worked



Checked DynamoDB for confirmation 1 item showing



In conclusion I have set up a kinesis datastream and set up a role with policies to have access to dynamodb and kinesis, then I created a Lambda function an attached

created role. I then went on to Create a DynamoDB database to which my lambda function python code would paste all items sourced from the kinesis data stream, created an event which has worked